SQL Feature Summary Report

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• Indexes:

o Function1: movieSearch
Description: A nonclustered index on Title in the Movies table is used when the user searches by movie name. It speeds up the global search bar and the stored procedure that searches movies by title.

o Function2: topRatedMovies
Description: Nonclustered indexes on VoteAverage and VoteCount in Movies support the Top Rated Movies feature by making ORDER BY rating and popularity efficient when returning the Top 100 movies.

o Function3: genreBrowseAndRecommendations
Description: Indexes on GenreID and MovieID in MovieGenres optimise joins between Movies, Genres and MovieGenres. They are used when filtering by genre and when recommending movies based on a user's favourite genres.

o Function4: userActivityAndFeeds
Description: Indexes on (MovieID, ReviewDate) and UserID in Reviews, plus indexes on UserID in Watchlist and ListID in ListItem, speed up user activity, watchlist, custom list and trending queries that read recent rows.

• Triggers:

o Function1: syncMovieRatings
Description: The trigger trigger_UpdateMovieRating fires after changes on Reviews and recalculates VoteAverage and VoteCount in Movies, keeping rating aggregates correct for Top Rated and recommendation features.

o Function2: autoWatchlistCleanup
Description: The trigger trgger_CleanupWatchlist runs after a new review is inserted and deletes the same (UserID, MovieID)from Watchlist, so movies are automatically removed from watchlists once they are reviewed.

o Function3: reviewDateValidation
Description: A validation trigger on Reviews is designed to prevent invalid or future-dated review timestamps and works together with the proced_AddReview procedure, which always sets ReviewDate using GETDATE().

• Stored Procedures:

o Function1: topRatedMovies
Description: proced_GetTopRatedMovies returns the top 100 movies ordered by VoteAverage and VoteCount and is used by the Top Rated Movies screen to fetch a ready-made list with a single database call.

o Function2: movieDetails
Description: proced_GetMovieById, proced_GetMovieGenres and proced_GetMovieReviews provide movie info, its genres and its latest reviews. The backend combines these into a MovieDetailsDTO for the movie details page.

o Function3: genreBrowseAndRecommend
Description: sp_GetMoviesByGenre returns movies for a selected genre (with a configurable limit), and proced_RecommendMovies returns recommended movies based on the user's top-reviewed genres. These support genre browsing and personalised recommendations.

o Function4: userActivityAndFeeds
Description: proced_GetUserActivity merges reviews, watchlist entries and list items into one ordered activity stream, proced_GetPublicActivityFeed and proced_GetFriendsActivityFeed return public and friends' review feeds, and proced_GetTrendingMovies returns movies with the highest recent review volume.

o Function5: socialGraphAndSuperUsers
Description: proced_FollowUser inserts follow relationships safely, proced_FindFriendConnections discovers friend-of-friend connections up to three levels, and proced_GetSuperUsers finds users whose review counts are above average for analytics.

o Function6: watchlistsAndLists
Description: proced_AddToWatchlist, proced_RemoveFromWatchlist and proced_GetWatchlist manage watchlist entries, while proced_GetUserLists, proced_CreateList, proced_AddListItem, proced_RemoveListItem and proced_GetListItems manage user-created lists and their movies.